AKKA

Scale AI, Efficiently
The Agentic Systems Platform

# Akka vs. LlamaIndex

A comparison for teams building agentic AI

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**LlamaIndex indexes and retrieves your data; Akka runs the agentic system that acts on it — and guarantees it.** LlamaIndex is a strong RAG / data framework: connectors, parsing, indexing, and retrieval that give a model the right context. It is not an agentic runtime — production reliability, durable state, HA/DR, and runtime governance are yours to build, integrate, and operate. The Akka Agentic AI Platform delivers them and guarantees resilience and scalability, and LlamaIndex's retrieval layer can feed an Akka agent.

| 99.9% | 99.9999% | 4ms | Up to 90% |
|---|---|---|---|
| LLAMAPARSE STATED UPTIME | AKKA AGENTIC AI PLATFORM SLA | AKKA DURABLE-STATE READS | CHEAPER TO OPERATE |

| DIMENSION | LLAMAINDEX | AKKA |
|---|---|---|
| **What it is** | A RAG / data framework for indexing and retrieval (plus LlamaParse, formerly LlamaCloud, a managed parsing, extraction, and indexing service) | The Akka Agentic AI Platform, which guarantees resilience and scalability |
| **Primary job** | Connect, parse, index, and retrieve your data so an LLM has the right context | Run, scale, persist, and govern agentic systems in production |
| **Agentic features** | Workflows (event-driven orchestration), function-calling and ReAct agents, AgentWorkflow — a library you deploy and operate, or deploy to LlamaParse with `llamactl` | Native agents, durable memory, streaming, APIs, orchestration, and governance on one runtime |
| **Durable execution** | Not built in — Workflows do not auto-checkpoint; durability needs an external integration (e.g., DBOS), replay-based and at-least-once | Durable sharded in-memory state, event-sourced, replayable; 4ms reads / sub-10ms writes |
| **Availability SLA** | 99.9% uptime stated for the LlamaParse service, terms negotiated per Enterprise agreement; no SLA on customer agents | 99.9999% — entire platform, backed by indemnities |
| **HA/DR** | Customer-owned (self-hosted) or per managed-service terms; no published active-active / RTO / RPO | Active-active across regions; sub-1-minute RTO; zero-byte RPO |
| **Governance / EU AI Act** | Observability and evaluation integrations; no inline enforcement, immutable ledger, pre-deployment classification, or sealed artifact | Aspect-woven runtime enforcement + full pre-production governance |
| **Cost model** | Consumption credits ($1.25 / 1,000 credits; basic parsing from 1 credit/page; 10,000 free credits/mo) + you provision and operate everything else | Shared compute; up to 90% lower infrastructure for the same workload, fixed annual fee |
| **Certifications** | SOC 2 Type II audit completed for LlamaParse; HIPAA BAA for Enterprise | 19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF) |
| **Vendor model** | Venture-funded: $19M Series A (Mar 2025), ~$93M post-money; Databricks & KPMG strategic minority investments | Profitable and self-funding; since 2009, 100,000+ deployments; Dell Technologies Capital is largest shareholder, customer, and AI partner |

## LlamaIndex Is a Retrieval Framework; Akka Is the Agentic Platform

LlamaIndex is a data framework for retrieval-augmented generation: data connectors, document parsing (LlamaParse), indexing, and query/retrieval interfaces that give a model the right context. It does this well, and it has added agentic constructs — Workflows (event-driven orchestration), function-calling and ReAct agents, and AgentWorkflow. Those are libraries you import, deploy, and operate. The substrate a production agentic system runs on — a durable runtime, high availability, disaster recovery, and embedded governance — is not part of LlamaIndex; you build, integrate, and operate it yourself, and own every failure across the seams.

Everything an agentic system needs runs in one Akka runtime, so a system that compiles is ready to run in production. The Akka SDK models the entire agentic system in one place — agents, tools, durable memory, streaming, APIs, and governance — and the platform generates it, tests it, and runs it as one certified service.

| Capability | LlamaIndex | Akka |
|---|---|---|
| **Data ingestion, parsing, indexing, retrieval (RAG)** | Yes — its core strength | Not native (see complement note) |
| **Agent constructs (tools, function calling, ReAct)** | Library | Built in |
| **Event-driven orchestration (Workflows)** | Library | Built in |
| **Durable execution / crash recovery** | External integration (e.g., DBOS) | Built into the runtime |
| **Durable memory** | Bolt-on store | Built in, 4ms / sub-10ms |
| **Real-time streaming** | None native | Built in, backpressured, petabyte-scale |
| **HTTP / gRPC API layer** | Workflows served as HTTP endpoints (WorkflowServer, hosted deploy) | Built in |
| **Runtime governance / policy enforcement** | None | Inline, runtime-embedded |
| **Pre-production governance** | None | Classification, sign-offs, sealed posture |

## Availability, Durability, and Disaster Recovery

LlamaIndex states 99.9% uptime for the LlamaParse service, with contractual uptime commitments negotiated per Enterprise agreement and no published RTO or RPO; no SLA covers the agents a customer runs. More important for agentic AI is durable execution. LlamaIndex Workflows do not automatically snapshot state — by design, to avoid overhead — so a workflow cannot recover from a crash on its own. Durable execution is available only through an external integration such as DBOS, which journals step completions to a database and replays on restart; recovery is at-least-once, so steps can run more than once and the developer must make them idempotent.

| Metric | LlamaIndex | Akka |
|---|---|---|
| **Availability SLA** | 99.9% stated for LlamaParse; none for customer agents | 99.9999% |
| **Allowed downtime / year** | Not specified | ~31 seconds |
| **Durable state** | External integration; replay-based, at-least-once | Event-sourced, built in; exactly-once recovery |
| **State latency** | Per external store | 4ms reads / sub-10ms writes |
| **RTO / RPO** | Not published | Sub-1-minute RTO / zero-byte RPO |
| **HA/DR** | Customer-owned / per managed terms | Active-active across regions |
| **SLA scope** | — | The entire platform, backed by indemnities |

Akka provides durability and fault tolerance as part of the runtime: state is event-sourced in durable sharded in-memory storage, replayable from its own journal, with active-active HA/DR, sub-1-minute RTO, and zero-byte RPO under a 99.9999% SLA. 100,000+ production deployments since 2009 stand behind it.

## Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

## Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. The drivers are actor concurrency (~10 trillion tokens/core/year vs ~2 trillion for comparable solutions; ~80% less compute than Python-based frameworks), shared compute, and micro-checkpointing. Manulife reported up to 300% more concurrency and 30–50% faster processing after porting Python-based systems to Akka.

LlamaParse, formerly LlamaCloud, bills by consumption — 1,000 credits cost $1.25, basic parsing starts at 1 credit per page, and the free plan includes 10,000 credits per month — and that meter covers parsing, extraction, and indexing: the retrieval layer. The runtime to run agents in production (compute, memory, streaming, APIs, observability, governance, HA/DR) is provisioned and paid for separately, on top. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast.

## Governance and the EU AI Act

LlamaIndex provides observability and evaluation: instrumentation, tracing integrations, and evaluation tooling to measure retrieval and agent quality. Agent Workflows can pause for human review at steps developers write into the workflow. LlamaIndex does not provide real-time policy enforcement, an immutable interaction ledger, pre-deployment classification, or a sealed audit artifact, and publishes no mapping to the EU AI Act.

| Violation | Maximum Fine |
|---|---|
| Prohibited AI practices (Art. 5) | EUR 35M or 7% global turnover |
| High-risk obligations (Art. 9-15) | EUR 15M or 3% global turnover |

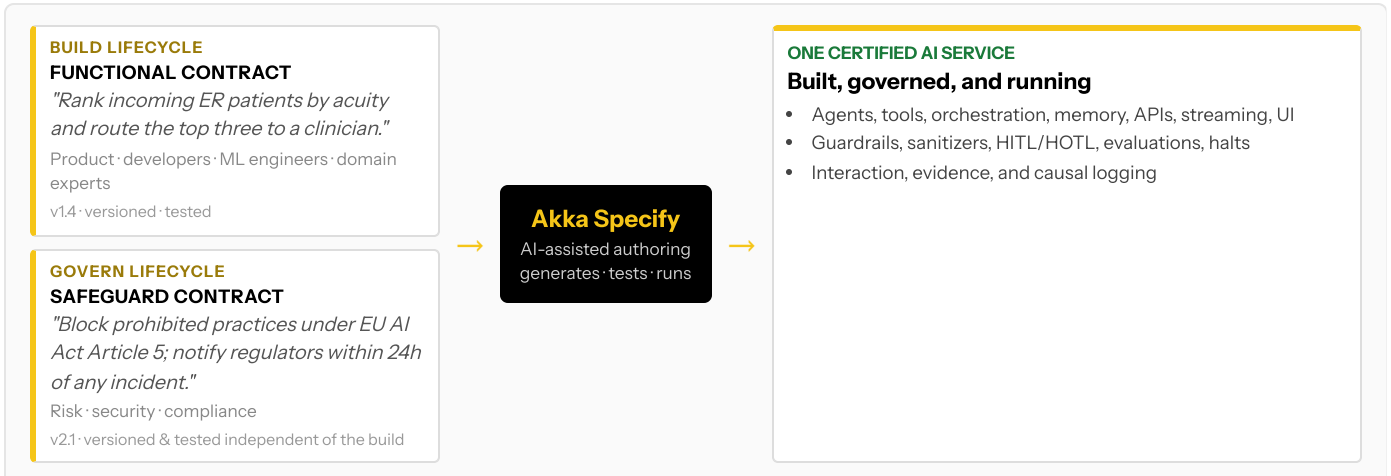
High-risk AI carries a 10-year logging-retention obligation (Art. 72), enforceable since Feb 2025 (prohibited) / Aug 2025 (high-risk).

### How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL human control; atomic PII scrub-with-explain; pre-deployment classification against **241 regulations and frameworks and 1,718 controls** (699 carrying a financial penalty); multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance that LlamaIndex would leave a customer to source and bolt on, Akka enforces inline.

## Two Lifecycles, One Certified System

Building on LlamaIndex means engineers writing retrieval and workflow code, or generating extraction workflows from a plain-language description in LlamaAgents Builder (beta); there is no built-in path for a risk officer or compliance lead to contribute, and no governance lifecycle. Akka runs two independent lifecycles on one platform via **Akka Specify**:



**Akka Verify** ↻ validates the running system against both specs and fine-tunes the AI from production data.

The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow LlamaIndex has no equivalent for.

## Real-Time Streaming at Petabyte Scale

LlamaIndex has no streaming engine; real-time pipelines are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

## For the Buyer: Risk, Compliance, and Accountability

| Buyer concern | LlamaIndex | Akka |
|---|---|---|
| **Certifications & audits** | SOC 2 Type II audit completed for LlamaParse (formerly LlamaCloud); HIPAA BAA for Enterprise | 19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io) |
| **Scope of accountability** | The retrieval layer; you integrate and operate the runtime, agents, and governance | One platform, one SLA, 24/7 SRE — Akka owns the running system |
| **Risk transfer** | Standard cloud terms | Availability and data-integrity guarantees backed by contractual indemnities |
| **Track record & funding model** | Venture-funded: $19M Series A (Mar 2025), ~$93M post-money; Databricks and KPMG strategic minority investments | Profitable and self-funding; since 2009 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner |
| **Budget predictability** | Consumption credits that scale with load | Fixed annual fee finance can forecast |

The decision is scope and accountability: LlamaIndex gives you a best-in-class retrieval layer to feed an agent; Akka gives you the platform that runs and guarantees the agent.

> **Akka complements your retrieval layer.** Akka is not a vector database or a semantic knowledge layer, and does not aim to be. LlamaIndex's strength — parsing, indexing, and retrieving your data — is real and complementary. A LlamaIndex retrieval pipeline can supply context to an Akka agent, which then runs that agent as a durable, highly available, governed production system. Retrieval feeds the agent; Akka runs it.

## Customers Running Agentic and Real-Time Systems on Akka

**Manulife**
**2,000**
developers across 100 projects on one governed platform

**Tubi**
**5B tok/s**
real-time hyper-personalization engine

**Swiggy**
**71ms**
order-assignment AI, ~50% latency reduction

**John Deere**
**1,000+**
tractor sensors turned into real-time insight

**Verizon**
**750%**
order-processing capacity gain; 6s → 2.4s response

## Common Questions

### We already use LlamaIndex for RAG. Why add Akka?

LlamaIndex is strong for connecting, indexing, and retrieving your data. To put an agent into production you also need a durable runtime, high availability and disaster recovery, durable memory, streaming, and runtime governance — which LlamaIndex does not provide. Akka runs the agent as a guaranteed production system, and a LlamaIndex retrieval pipeline can feed it.

### LlamaIndex has Workflows and agents now. Isn't that an agentic runtime?

Workflows and agents are libraries you deploy and operate yourself, or deploy to LlamaParse with `llamactl`. They do not auto-checkpoint state, so crash recovery requires an external integration such as DBOS journaling to a database, with at-least-once semantics you design

### Can we add governance on top of LlamaIndex?

You can add observability and evaluation tools and code human review steps into Workflows, but the EU AI Act expects enforcement inline to the runtime: immutable records witnessed as they happen, human override on running processes, and pre-deployment classification. Tools that observe and evaluate cannot gate a deployment or seal an audit artifact. Akka embeds all of this and covers pre-deployment governance.

### Is LlamaIndex cheaper because the framework is open source?

The OSS framework is free, but production means LlamaParse consumption credits for parsing and indexing plus the separate runtime you provision and operate for everything else. Akka's shared-compute model is up to 90% cheaper to operate than the equivalent assembled stack, on a fixed annual fee.

around. Akka provides durable, event-sourced execution, active-active HA/DR, and a 99.9999% SLA as part of the runtime.

## Sources


**LlamaIndex product / RAG:** llamaindex.ai · developers.llamaindex.ai/python/framework/ — open-source data framework for RAG (ingest, index, retrieve); "AI Agents for Document OCR + Workflows" (Jun 2026)

**LlamaIndex agentic constructs:** developers.llamaindex.ai/python/llamaagents/workflows/ — Workflows (event-driven), AgentWorkflow, function-calling / ReAct agents

**Workflows durability:** developers.llamaindex.ai/python/llamaagents/workflows/dbos/ — Workflows do not auto-snapshot; durable execution via DBOS journaling, replay-based, at-least-once (idempotency required) (2026)

**LlamaParse pricing:** llamaindex.ai/pricing — 1,000 credits = $1.25; basic parsing from 1 credit/page; Free 10K credits/month, Starter $50 (40K), Pro $500 (400K), Enterprise custom; LlamaCloud renamed LlamaParse (llamaindex.ai/blog/llamaindex-newsletter-2026-02-24)

**LlamaIndex SLA / security:** llamaindex.ai/pricing; developers.llamaindex.ai/llamaparse/general/enterprise-readiness/ — "99.9% uptime" stated for the LlamaParse service; contractual commitments negotiated per Enterprise agreement; no published RTO/RPO; SOC 2 Type II audit, HIPAA BAA, EU DPA

**LlamaIndex funding:** prnewswire.com / crunchbase.com — $19M Series A led by Norwest with Greylock, Mar 4 2025, ~$93M post-money, $27.5M total disclosed; Databricks Ventures + KPMG strategic minority investments (May 2025)

**Akka trust center:** trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

**Akka performance:** akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

**Akka platform:** 99.9999% availability, active-active HA/DR, sub-1-min RTO, zero-byte RPO (contractual indemnities); 4ms reads / sub-10ms writes; 241 regulations and frameworks / 1,718 controls / 699 with a financial penalty; 100,000+ deployments / since 2009 / 52 banks; profitable; Dell Technologies Capital